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University ST4005CEM Database System

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1. Find the total billing amount paid by each patient. Display PatientID and total amount.

175

```
176 • select patientID,sum(amount) as total_amount from billing group by patientID;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	patientID	total_amount		
▶	2	1500.00		
	3	2300.50		
	5	3700.00		
	6	4750.25		
	101	9100.00		

2. Count the total number of appointments booked for each doctor. Display Doctorname and appointment count.

```
175 • select concat(d.firstName," ",d.lastName) as Doctors,count(a.appointmentID) as appointments
```

```
176 from doctor d join appointment a on d.doctorID=a.doctorID group by a.appointmentID;
```

177

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Doctors	appointments		
▶	Dr. raman khatiwada	1		
	Rajesh Karki	1		
	Anita Shrestha	1		
	Suresh Pandey	1		
	Mina Tamang	1		
	Dipak Adhikari	1		
	Rajesh Karki	1		

3. Find the average billing amount from the Billing table.

```
178 • select round(avg(amount),2) as average_bill_amount from billing;
```

179

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	average_bill_amount			
▶	2668.84			

4. Display the maximum and minimum billing amount recorded in the Billing table.

```
180 • select max(amount) as max_bill_amount from billing;
```

181

Result Grid		Filter Rows:	Export:	Wrap Cell Contents:
	max_bill_amount			
▶	5000.00			

```
180 • select min(amount) as max_bill_amount from billing;
```

181

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	max_bill_amount			
▶	500.00			

5. List all doctors who have issued more than 2 prescriptions. Display doctorname and prescription count.

```
182 • select concat(d.firstName," ",d.lastName) as Doctors, count(p.prescriptionID)
183 as prescription_count from prescription p join doctor d on d.doctorID=p.doctorID
184 group by p.doctorID having count(p.prescriptionID) >2 ;
185
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Doctors	prescription_count		

6. Find the total number of medical records grouped by Diagnosis. Display Diagnosis and record count.

```
199 • select diagnosis, count(diagnosis) as number_of_patients
200 from medicalRecord group by (diagnosis);
201
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	diagnosis	number_of_patients			
▶	Hypertension	2			
	Diabetes	2			
	Asthma	1			
	Flu	1			

7. Display the full name of each patient along with their diagnosis from the MedicalRecord table.

```
203 • select concat(p.firstName, " ", p.lastName) as patientName, m.diagnosis
204 from medicalRecord m join patient p on p.patientID=m.patientID;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	patientName	diagnosis			
▶	Ram Sharma	Hypertension			
	Priya Thapa	Diabetes			
	Bikash Rai	Hypertension			
	Sita Gurung	Asthma			
	Rohan Magar	Diabetes			
	Aarav Sharma	Flu			

8. List all appointments along with the patient's first name, last name, and the doctor's first name and last name.

```

206
207 • select a.appointmentID,concat_ws(" ",p.firstName,p.lastName,p.patientID) as PatientName,
208      concat(d.firstName, " ",d.lastName," ",d.doctorID) as DoctorName, a.reason,a.status
209      from appointment a join patient p on p.patientID=a.patientID
210      join doctor d on d.doctorID=a.doctorID ;

```

appointmentID	PatientName	DoctorName	reason	status
1	Ram Sharma 101	Dr. raman khatiwada 1	All good	COMPLETED
2	Ram Sharma 101	Rajesh Karki 2	Chest pain checkup	Completed
11	Rohan Magar 5	Rajesh Karki 2	Heart rate monitoring	Scheduled
3	Priya Thapa 2	Anita Shrestha 3	Headache and dizziness	Completed
4	Bikash Rai 3	Suresh Pandey 4	Knee joint pain	Completed
5	Sita Gurung 4	Mina Tamang 5	Child fever	Completed
6	Rohan Magar 5	Dipak Adhikari 6	Skin rash treatment	Scheduled

9. Show all prescriptions along with the medication name, patient name, and the prescribing doctor's name.

```

213 • select concat_ws(" ",p.prescriptionID, p.medication,p.dosage,p.startDate,p.endDate )
214      as prescriptions , concat(pt.firstName, " ",pt.lastName) as patientName,
215      concat(d.firstName, " ",d.lastName) as DoctorName
216      from prescription p join patient pt on pt.patientID= p.patientID
217      join doctor d on d.doctorID=p.doctorID ;
218

```

prescriptions	patientName	DoctorName
13 Amoxicillin 500mg twice 2025-01-01 2025-02-10	Ram Sharma	Dr. raman khatiwada
18 Paracetamol 650mg thrice 2025-02-01 2025-03-12	Aarav Sharma	Dr. raman khatiwada
17 Cetirizine 10mg once 2025-01-20 2025-03-01	Rohan Magar	Dr. raman khatiwada
16 Atorvastatin 20mg once 2025-01-15 2025-03-16	Sita Gurung	Dr. raman khatiwada
15 Metformin 850mg twice 2025-01-10 2025-03-21	Bikash Rai	Dr. raman khatiwada
19 Painkiller 50 mg twice 2026-05-13 2026-06-02	Priya Thapa	Dr. raman khatiwada
14 Ibuprofen 400mg thrice 2025-01-05 2025-02-14	Priya Thapa	Dr. raman khatiwada
13 Amoxicillin 500mg twice 2025-01-01 2025-02-10	Ram Sharma	Rajesh Karki
18 Paracetamol 650mg thrice 2025-02-01 2025-03-12	Aarav Sharma	Rajesh Karki
17 Cetirizine 10mg once 2025-01-20 2025-03-01	Rohan Magar	Rajesh Karki
16 Atorvastatin 20mg once 2025-01-15 2025-03-16	Sita Gurung	Rajesh Karki
15 Metformin 850mg twice 2025-01-10 2025-03-21	Bikash Rai	Rajesh Karki
19 Painkiller 50 mg twice 2026-05-13 2026-06-02	Priya Thapa	Rajesh Karki

10. List all patients and their billing details. Include patients who have no billing records as well.

```

220 • select * from patient p left join billing b on p.patientID=b.patientID ;
221

```

patientID	firstName	lastName	dateOfBirth	gender	Phone	BloodGroup	billingID	patientID	amount	billindDate	status
2	Priya	Thapa	1985-07-22	female	9812345678	NULL	2	2	1500.00	2025-01-11	Completed
3	Bikash	Rai	2000-11-08	male	9823456789	NULL	3	3	2300.50	2025-01-16	Completed
4	Sita	Gurung	1995-01-30	female	9834567890	NULL	NULL	NULL	NULL	NULL	NULL
5	Rohan	Magar	1978-06-12	others	9845678901	NULL	5	5	3200.00	2025-02-02	Completed
5	Rohan	Magar	1978-06-12	others	9845678901	NULL	10	5	500.00	2025-03-16	Pending
6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	6	6	950.25	2025-02-11	Pending
6	Aarav	Sharma	1990-03-15	male	9801234567	NULL	11	6	3800.00	2025-03-21	Pending
101	Ram	Sharma	1990-05-10	male	9812345678	NULL	1	101	5000.00	2026-05-08	Pending
101	Ram	Sharma	1990-05-10	male	9812345678	NULL	7	101	4100.00	2025-02-21	Completed

11. Display all doctors and any appointments they have. Show doctors with no appointments too.

```

224 • select * from doctor d left join appointment a on d.doctorID=a.doctorID;
225

```

doctorID	firstName	lastName	speciality	phone	email	appointmentID	patientID	doctorID	appointmentDate	reason
1	Dr. raman	khatriwada	Cardiology	9812311222	ramankhatriwada@hospital.com	1	101	1	2026-05-08	All good
2	Rajesh	Karki	Cardiology	9851234567	rajesh@gmail.com	2	101	2	2025-01-10	Chest pain che
2	Rajesh	Karki	Cardiology	9851234567	rajesh@gmail.com	11	5	2	2025-03-20	Heart rate mon
3	Anita	Shrestha	Neurology	9862345678	anita@gmail.com	NULL	NULL	NULL	NULL	NULL
4	Suresh	Pandey	Orthopedics	9873456789	suresh@gmail.com	4	3	4	2025-01-20	Knee joint pain
5	Mina	Tamang	Pediatrics	9884567890	mina@gmail.com	5	4	5	2025-02-01	Child fever
6	Dipak	Adhikari	Dermatology	9895678901	dipak@gmail.com	6	5	6	2025-02-10	Skin rash treat

12. Find each patient's full name and the total amount billed to them by joining Patient and Billing tables.

```

225
226 • select concat(p.firstname, " ",p.lastName) as "Patient Name", sum(b.amount) as "Total cost"
227 |from patient p join billing b on p.patientID=b.patientID group by (p.patientID);
228

```

Patient Name	Total cost
Priya Thapa	1500.00
Bikash Rai	2300.50
Rohan Magar	3700.00
Aarav Sharma	4750.25
Ram Sharma	9100.00

13. List appointment details (date, reason, status) together with the patient's gender and contact info.

```
228
229 • select concat_ws(" ", a.appointmentdate,a.reason,a.status) as "Date    Reason    Status ",
230    concat_ws(" ",p.gender,p.phone) as "Gender    Contact"
231 from appointment a join patient p on a.patientID=p.patientID;
```

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content:
	Date	Reason	Status	Gender	Contact
▶	2026-05-08	All good	COMPLETED	male	9812345678
	2025-01-10	Chest pain checkup	Completed	male	9812345678
	2025-01-20	Knee joint pain	Completed	male	9823456789
	2025-02-01	Child fever	Completed	female	9834567890
	2025-02-10	Skin rash treatment	Scheduled	others	9845678901
	2025-03-20	Heart rate monitoring	Scheduled	others	9845678901

14. Retrieve all appointments where the status is 'Scheduled' and appointments is after Jan 1 2026.

```
234 • select * from appointment a where appointmentDate > "2026-01-01" and status ="scheduled";
```

[illegible]

15. Find all patients whose first name starts with the letter 'A'.

```
236 • select * from patient where firstName like "a%";
```

[illegible]

16. Display all billing records where the amount is between 500 and 2000.

```
238 • select * from billing where amount between 500 and 2000;
```

	billingID	patientID	amount	billindDate	status
▶	2	2	1500.00	2025-01-11	Completed
	6	6	950.25	2025-02-11	Pending
	10	5	500.00	2025-03-16	Pending
*	NULL	NULL	NULL	NULL	NULL

17. List all prescriptions where the medication is either 'Paracetamol', 'Ibuprofen', or 'Amoxicillin'.

```
240 • select * from prescription where medication in ( 'Paracetamol', 'Ibuprofen', 'Amoxicillin' ) ;
```

	prescriptionID	patientID	doctorID	medication	dosage	startDate	endDate
▶	13	101	1	Amoxicillin	500mg twice	2025-01-01	2025-02-10
	14	2	2	Ibuprofen	400mg thrice	2025-01-05	2025-02-14
	18	6	3	Paracetamol	650mg thrice	2025-02-01	2025-03-12
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

18. Retrieve medical records where the treatment contains the word 'Surgery' or the notes are not null.

```
242 • select * from medicalrecord where treatment like "%surgery%" or treatment is not null;
```

	recordID	patientID	diagnosis	treatment	recordDate	Notes
▶	1	101	Hypertension	Medication	2026-05-08	Follow the instruction
	9	2	Diabetes	Insulin Therapy	2026-05-13	Monitor blood sugar daily
	10	3	Hypertension	Medication	2026-05-13	Reduce salt intake
	11	4	Asthma	Inhaler	2026-05-13	Avoid dust and smoke
	12	5	Diabetes	Oral Medication	2026-05-13	Check HbA1c monthly
	13	6	Flu	Rest and Fluids	2026-05-13	Stay hydrated
*	NULL	NULL	NULL	NULL	NULL	NULL

19. Add a new column called BloodGroup of type VARCHAR(5) to the Patient table.

```

243 • select * from patient;
244 • desc patient;
245 • alter table patient add column BloodGroup varchar(5);

```

Field	Type	Null	Key	Default	Extra
patientID	int	NO	PRI	NULL	
firstName	varchar(20)	YES		NULL	
lastName	varchar(20)	YES		NULL	
dateOfBirth	date	YES		NULL	
gender	enum('male','female','others')	YES		NULL	
Phone	varchar(15)	YES		NULL	
BloodGroup	varchar(5)	YES		NULL	

Result 67 x Reac

Output

#	Time	Action	Message
82	16:55:38	desc patient	7 row(s) returned
83	16:56:20	alter table patient add column BloodGroup varchar(5)	Error Code: 1060. Duplicate column name 'BloodGroup'
84	16:56:27	desc patient	7 row(s) returned

20. Modify the Phone column in the Doctor table to change its data type from VARCHAR(15) to VARCHAR(20).

```

247 • alter table doctor modify column phone varchar(20);
248 • desc doctor;

```

Result Grid						
Filter Rows:						
Export:						
Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	doctorID	int	NO	PRI	NULL	
	firstName	varchar(20)	YES		NULL	
	lastName	varchar(20)	YES		NULL	
	speciality	varchar(20)	YES		NULL	
	phone	varchar(20)	YES		NULL	
	email	varchar(50)	YES		NULL	